

Lupita Vásquez Fabela

New York City ▪ gsvfabela@gmail.com ▪ lupitavsqf.github.io/Portfolio

Systems-focused AI product builder working at the intersection of cognition, interaction design, and large language models. Designs and prototypes that prioritize accessibility and human skill development for a collective benefit.

Experience

Founder / Product Manager and Interaction Designer

August 2024 – Present

 **BIG FROOTS — Inclusive EdTech**

New York City

Incubated during the Impact Entrepreneurship Initiative Fellowship at Parsons School of Design

[BIG FROOTS website](https://bigfroots.com)

- Designed and prototyped agentic AI systems with LLMs centered on accessibility, using cognitive constraints to reduce computational costs, translating research into documented, reproducible workflows — including the Eleva middleware layer, a working memory orchestration system for LLM-assisted coding.
- Developed patent-pending lightweight orchestration layers, databases, and data pipelines for LLM powered agents, optimizing short-term memory (or context), tool-calling behavior, and recovery from non-deterministic failure paths.
- Conducted user research on cognitive load and non-deterministic AI behavior, producing educational frameworks for how developers and end users interact with LLM agents.
- Developed prototypes of digital equestrian gear with haptics for accessibility in horse-riding therapy sessions.
- Built **CyberCowboys**, an augmented-reality riding-lesson system for Snap Spectacles: a real-time web app and course designer (Node.js, SQLite) synced over WebSockets to a Lens Studio (TypeScript) AR lens that projects courses, obstacles, and paths onto the real arena floor with live floor detection and calibration.
- Developing prototypes of digital assistive technologies (XR, AR) for equestrian sports with Snap Spectacles.

IT Transition and Delivery Manager

March - October 2023

Nissan

Mexico City

IT Governance and PMO

- Oversaw 7 project managers across 23 concurrent initiatives within IT Governance and PMO operations.
- Designed automated data pipelines and reporting workflows in ServiceNow, reducing manual reporting overhead and automating data visualization, adhering to Kanban and Scrum frameworks.
- Designed and automated PowerBI dashboards to support executive-level visibility into delivery timelines, risk, and resource allocation.

Program Manager

November 2022 – March 2023

Blue People

Mexico City

Business and Software Development

- Led the end-to-end development of an AWS-based e-commerce platform supporting logistics and warehouse optimization.
- Consolidated infrastructure into a single returns-handling facility by redesigning backend workflows and operational data flows.
- Acted as Scrum Master for AWS-based software delivery teams, facilitating Agile ceremonies, sprint execution, and cross-functional coordination between engineering, product, and operations.
- Designed data pipelines integrating Kanban systems into sales management portals, improving data visualization and cross-team coordination.

Business Development Lead

December 2020 – October 2022

eJAmerica

California, USA

Business Development and Strategic Operations

- Designed and executed data-driven lead-generation campaigns, contributing to 40% of new client acquisition during 2021–2022.
- Facilitated Agile workflows between sales, product, and engineering teams; led sprint planning and backlog alignment to translate client RFP requirements into executable technical deliverables.
- Led technical presentations and stakeholder meetings, facilitating alignment between business objectives and engineering capabilities.
- Conducted market research to assess strategic initiatives, contributing to website renewal, enhancement of the in-house SaaS recruitment platform (eJRekrut), and digital communication strategy.

Education

MFA Design and Technology

2024 – 2026

Parsons School of Design

New York City

Merit Scholarship by The New School & Full Scholarship by SECIHTI, Mexican Government

Impact Entrepreneurship Initiative Fellow

B.Sc. Chemistry

2022

Facultad de Química, Universidad Nacional Autónoma de México

Mexico City

Specialization in Computational Chemistry and Philosophy of Chemistry

Skills

AI Systems & Orchestration: Agentic systems, LLM orchestration, short-term memory optimization, non-deterministic workflows

Data & Modeling: Statistical analysis, machine learning, data visualization, metrology

Programming: Python, SQL, C++, HTML, CSS, JavaScript ecosystem (Node.js, React, TypeScript)

ML Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib

Agile Leadership: Scrum (multi-team), Kanban, Sprint Planning, Retrospectives, Backlog Grooming, Risk Mitigation

Product & Workflow: Jira, Confluence, Azure DevOps, Git, ServiceNow, PowerBI

Design & Prototyping: Figma, Adobe XD, Illustrator, Premiere Pro, After Effects

Languages: Spanish (Native), English (Fluent), French (B1), Portuguese (A1)

Certifications

HCIA-AI

Huawei Training Center, Hangzhou, China, 2024

AWS Partner Accreditation (Technical)

Amazon Web Services, 2022

ITIL Foundation Level

PeopleCert, 2022

Professional Scrum Master I (PSM I)

Scrum.org, 2022

Awards and Honors

Invited Reviewer, NeurIPS Creative AI Track

Neural Information Processing Systems, 2026

base44 x Wix Cross Campus Hackathon - The New School and NYU (Second Place)

Wix

Playground. New York City, April 2026

PlayfulNYC Creative Technology Hackathon (Grand Prize and Best of Hardware) *The New School Innovation Center. New York City, April 2026*

Impact Entrepreneurship Initiative Fellowship (10K USD towards tuition)

The New School, New

York City, 2024

Parsons School of Design - Merit Scholarship (90K USD towards tuition)

The New School, New

York City, 2024

SECIHTI - Mexican Government Scholarship for Graduate Studies Abroad (53K USD towards tuition and living expenses) *Mexican Government, Mexico City, 2024*

IA 1000, Top 100 HCIA-AI Certification Scores in Mexico

Mexico City and Hangzhou, China, 2024

Premio Impacto a la Innovación UNAM, Second Place

Mexico City, 2019

National Chemistry Olympiad, Silver Medal

Guadalajara, México, 2012

Publications and Preprints

A Pragmatist Approach for New Dynamic Ontologies in Chemistry

2022

Undergraduate Thesis, Universidad Nacional Autónoma de México

Supervised by M.Sc. Luis Avelino Sánchez Graillet

Explored temporality as a core dimension of chemical identity, challenging reductionist models and laying foundations for pluralistic dynamic system representation in computational contexts.

[Spanish version](#) [English version](#)

Philosophical Pragmatism and Design as a Workaround to Common Fallacies in Executive Function Enhancement through Artificial Intelligence

2025

Analyzes limitations of reductionist AI narratives through design and pragmatist philosophy, with implications for cognition-aware AI systems.

[English version](#)

ReinGuide — Smart Haptic Reins for Inclusive Horse-Riding Therapy

2024–2025

MFA Thesis, Parsons School of Design · BIG FROOTS

Designed and evaluated a haptic interface for equine-assisted therapy, translating verbal riding instructions into low-intensity vibrotactile cues for nonverbal and autistic children.

Covers materials research, embedded prototyping, and user testing across therapy sessions, with findings indicating reduced cognitive load and improved participant engagement.

CyberCowboys — Augmented-Reality Riding Lessons for Snap Spectacles

2025–2026

BIG FROOTS

Designed and built a two-part AR system pairing a live course-design web app with a Snap Spectacles lens that projects riding-lesson paths, obstacles, and waypoints directly onto the arena floor.

Implements real-time WebSocket synchronization, floor detection, and arena calibration so instructors and riders share a single augmented view of the lesson.

Designing a Cognitively Accessible Coding Assistant with Memory-Aware LLM Orchestration

2026

Research Proposal, Parsons · UNAM

Introduces a structured memory orchestration (SMO) layer for LLM-based coding assistants, replacing full-history context injection with a bounded state representation to reduce token growth from $O(n^2)$ to near-linear scaling.

Applies chunking and progressive disclosure to support novice programmers, with a within-subjects study ($n = 18\text{--}35$) evaluating efficiency, comprehension, and cognitive load. Currently in pilot phase.

The Most Accurate Fortune Teller: An AI-Enhanced Pseudoscientific Toy As a Thought Experiment

2025

NeurIPS 2025 Creative AI Track, rejected

An AI-enhanced fortune-telling toy using biometric input (heart rate) and generated text to critically examine how technological mystique and emotional cues can create a false impression of algorithmic insight.

[View research portfolio](#)